

99acres Projects Module Automation Testing Report

Selenium Automation Testing using Pytest and BDD Behave Frameworks

Project Title

Selenium Automation Testing of 99acres Projects Module Using Pytest and BDD Behave Frameworks

Project Description

This project focuses on automation testing of the 99acres Projects module using Selenium WebDriver with Python. The framework validates the Projects property search workflow, location search, suggestions, filter application, project detail navigation, contact form handling, and View Number validation.

The project contains two automation implementations: Pytest for direct test automation and Behave BDD for business-readable feature scenarios. Screenshots, logs, and Allure reports are generated to support test evidence and debugging.

Test Strategy

Functional Testing

UI Testing

Positive Testing

Negative Testing

Regression Testing

End-to-End Testing

BDD Scenario Testing

Test Type

End-to-End Testing, Positive Testing, Negative Testing, Smoke Testing, and Regression Testing.

Preconditions

User should have a valid internet connection.

Chrome browser should be installed.

Selenium WebDriver environment should be configured.

Python, Pytest, Behave, and Allure dependencies should be installed.

Valid mobile number should be available for OTP login when login is enabled.

Test configuration should be available in config.ini.

Test Data

Field	Value
Application URL	https://www.99acres.com/
Module	Projects
Location	Mumbai
Property Type	Flat/Apartment
Bedroom	3 BHK
Possession In Timeline	2036

Test Steps

Step No.	Action	Expected Result
1	Launch browser	Browser launches successfully
2	Open 99acres homepage	Homepage loads successfully
4	Open Projects module	Projects search page opens
5	Select configured location	Mumbai is selected
6	Submit Projects search	Projects results are displayed
7	Apply property type filter	Flat/Apartment filter is applied
9	Apply bedroom filter	3 BHK filter is applied
10	Open 3 rd project from results	Project detail page opens
11	Close disclaimer if visible	Disclaimer popup closes
12	Click View Number	Contact form or contact details appear
13	Fill contact form details	Name, phone/email fields are completed if shown
Step No.	Action	Expected Result
14	Possession In 2036 if asked	Buying timeline is selected
15	Click final View Number	Contact details or contact form state is validated

Expected Result

User should successfully access the 99acres homepage.

Projects module should open successfully.

Valid location search should display Projects project results. Property type, Possession In, and bedroom filters should apply correctly. Project detail page should open from the result list.

View Number flow should display contact details or contact form.

Contact form should support filling user details and selecting 2036 when the buying timeline question appears.

Actual Result

Projects module opened successfully. Mumbai

location search executed successfully. Projects search results loaded correctly.

Filters were applied and results remained visible.

Project detail page opened successfully.

View Number contact form handling was automated for Pytest and BDD page objects.

Pytest end-to-end execution passed during validation.

Status

Passed

Pytest Automation Testing Report

Positive Test Cases with Details

Test Case	Description
test_open_projects_module_from_homepage	Verifies that the Projects module opens from the homepage and the search box is visible and empty.
test_search_valid_location_in_projects	Searches for the configured valid location and verifies that Projects results are displayed.
Test Case	Description
test_location_suggestions_display_for_partial_location	Enters partial location text and validates that Mumbai suggestions are shown.
test_residential_property_type_filter_applies_successfully	Applies the Flat/Apartment property type filter and validates that results remain loaded.

Negative Test Cases with Details

Test Case	Description
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test_invalid_location_does_not_show_valid_suggestions_or_proceed	Enters an invalid location and verifies that valid suggestions are not shown and the application does not navigate away.
test_search_does_not_proceed_without_location	Clicks Search without entering a location and verifies that the page stays on the Projects search screen.

End-to-End Test Case

Test Case	Description
test_projects_end_to_end_flow	Validates login, projects search, filters, project detail opening, disclaimer handling, contact form handling, and View Number flow.

BDD Automation Testing Report

Feature Files Used

projects_end_to_end.feature

projects_positive_negative.feature

BDD Test Strategy

BDD testing was implemented using Feature, Scenario, Given, When, and Then format. The scenarios cover homepage access, login when enabled, Projects navigation, valid location search, filters, project detail navigation, View Number validation, invalid location behavior, and empty search handling.

BDD Positive Scenarios

Scenario	Detailed Explanation
Open the Projects module successfully from the homepage	Validates that the user can open the Projects module and see an empty search box.
Search for a valid location in the Projects module	Searches for the configured valid location and validates that results are displayed.
Scenario	Detailed Explanation
Display location suggestions for a valid partial location	Checks that partial text such as Noi displays Noida suggestions.
Apply the Residential Property Type filter on search results	Applies Flat/Apartment filter and verifies that filtered results remain loaded.

BDD Negative Scenarios

Scenario	Detailed Explanation
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Invalid location should not show valid suggestions or proceed	Validates invalid location handling and confirms the application does not navigate away.
Search should not proceed without entering a location	Validates empty search behavior and confirms the Projects search box remains visible.

BDD End-to-End Scenario

Scenario	Detailed Explanation
Apply Projects filters and open View Number popup	Validates the complete flow from homepage and login to Projects search, filters, project detail page, disclaimer, contact form, and View Number.

Test Environment

Item	Details
Application	99acres Website
Module	Projects
Automation Tool	Selenium WebDriver
Programming Language	Python
Testing Frameworks	Pytest and Behave BDD
Design Pattern	Page Object Model
Browser	Chrome
Operating System	Windows
IDE	VS Code / PyCharm
Reporting Tool	Allure Report
Project Path	C:\99acres-framework

Test Execution

Pytest executed test suites:

test_end_to_end_flow.py

test_positive_negative_projects.py BDD

executed feature files:

projects_end_to_end.feature

projects_positive_negative.feature

Test Output

Category	Count / Details
Pytest Test Files	2
Pytest Main Test Cases	7 including E2E, positive, and negative tests
BDD Feature Files	2
BDD Scenarios	7 total scenarios
Positive Scenarios	4
Negative Scenarios	2
E2E Scenario	1
Validated Functionalities	Projects navigation, location search, suggestions, filters, project detail page, contact form, View Number, screenshots, logs, and Allure report generation

Test Reports

The project uses Allure Report for Pytest and BDD test reporting.

Pytest Report Commands

```
cd C:\99acres-framework\SeleniumPytest\99acres
..\..\venv\Scripts\Activate.ps1
pytest tests\test_end_to_end_flow.py::test_projects_end_to_end_flow -s -v
--alluredir=allure-results
pytest tests\test_positive_negative_projects.py -s -v --alluredir=allure-results
pytest tests -s -v --alluredir=allure-results
allure serve allure-results
```

BDD Report Commands

```
cd C:\99acres-framework
venv\Scripts\Activate.ps1
behave SeleniumBDD\99acres\features\projects_end_to_end.feature --no-capture
```

```
behave SeleniumBDD\99acres\features\projects_positive_negative.feature --no-capture
behave SeleniumBDD\99acres\features --no-capture
behave SeleniumBDD\99acres\features -f allure_behave.formatter:AllureFormatter -o
SeleniumBDD\99acres\allure-results
allure serve SeleniumBDD\99acres\allure-results
```

The Allure report includes:

Test suite names

Test case and scenario execution status

Execution duration

Passed, failed, skipped, or broken test status

Screenshots captured during execution

Logs and stack traces

Environment and executor information

Test history and trends where available

Conclusion

The Selenium automation framework for the 99acres Projects module successfully validates the main user journey and important positive and negative checks. Pytest supports direct automation execution, while BDD provides business-readable scenario coverage. The framework validates Projects search, filter application, project detail navigation, contact form handling, View Number workflow, screenshot capture, logs, and Allure reporting.